

OUTPUT SCREENSHOTS

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JupyterLab Python 3 (ipykernel)

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv("HR_comma_sep.csv")
df.head()
```

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years	sales	salary
0	0.38	0.53	2	157	3	0	1	0	sales	low
1	0.80	0.86	5	262	6	0	1	0	sales	medium
2	0.11	0.88	7	272	4	0	1	0	sales	medium
3	0.72	0.87	5	223	5	0	1	0	sales	low
4	0.37	0.52	2	159	3	0	1	0	sales	low

Data Quality Checking

```
[2]: df.isnull().sum()
```

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JupyterLab Python 3 (ipykernel)

```
7 promotion_last_5years 14999 non-null int64
8 sales 14999 non-null object
9 salary 14999 non-null object
dtypes: float64(2), int64(6), object(2)
memory usage: 1.1+ MB
```

```
[13]: df.shape
```

```
[13]: (14999, 10)
```

```
[14]: df.describe()
```

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years
count	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000
mean	0.612834	0.716102	3.803054	201.050337	3.498233	0.144610	0.238083	0.021268
std	0.248631	0.171169	1.232592	49.943099	1.460136	0.351719	0.425924	0.144281
min	0.090000	0.360000	2.000000	96.000000	2.000000	0.000000	0.000000	0.000000
25%	0.440000	0.560000	3.000000	156.000000	3.000000	0.000000	0.000000	0.000000
50%	0.640000	0.720000	4.000000	200.000000	3.000000	0.000000	0.000000	0.000000
75%	0.820000	0.870000	5.000000	245.000000	4.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	7.000000	310.000000	10.000000	1.000000	1.000000	1.000000

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JupyterLab Python 3 (ipykernel)

Performing EDA

```
[16]: corr=df.corr(numeric_only=True)
corr
```

```
[16]:
```

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years
satisfaction_level	1.000000	0.105021	-0.142970	-0.020048	-0.100866	0.058697	-0.388375	0.025605
last_evaluation	0.105021	1.000000	0.349333	0.339742	0.131591	-0.007104	0.006567	-0.008684
number_project	-0.142970	0.349333	1.000000	0.417211	0.196786	-0.004741	0.023787	-0.006064
average_monthly_hours	-0.020048	0.339742	0.417211	1.000000	0.127755	-0.010143	0.071287	-0.003544
time_spend_company	-0.100866	0.131591	0.196786	0.127755	1.000000	0.002120	0.144822	0.067433
Work_accident	0.058697	-0.007104	-0.004741	-0.010143	0.002120	1.000000	-0.154622	0.039245
left	-0.388375	0.006567	0.023787	0.071287	0.144822	-0.154622	1.000000	-0.061788
promotion_last_5years	0.025605	-0.008684	-0.006064	-0.003544	0.067433	0.039245	-0.061788	1.000000

```
[5]: sns.heatmap(corr,annot=True)
plt.show()
```

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JupyterLab Python 3 (ipykernel)

```
7 promotion_last_5years 14999 non-null int64
8 sales 14999 non-null object
9 salary 14999 non-null object
dtypes: float64(2), int64(6), object(2)
memory usage: 1.1+ MB
```

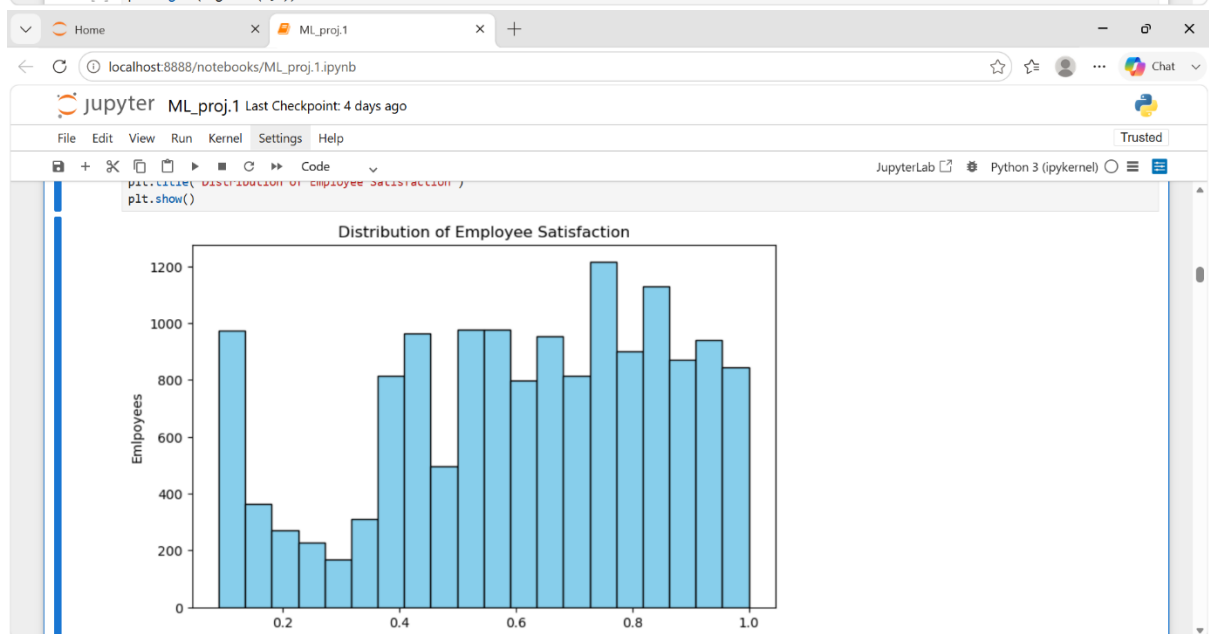
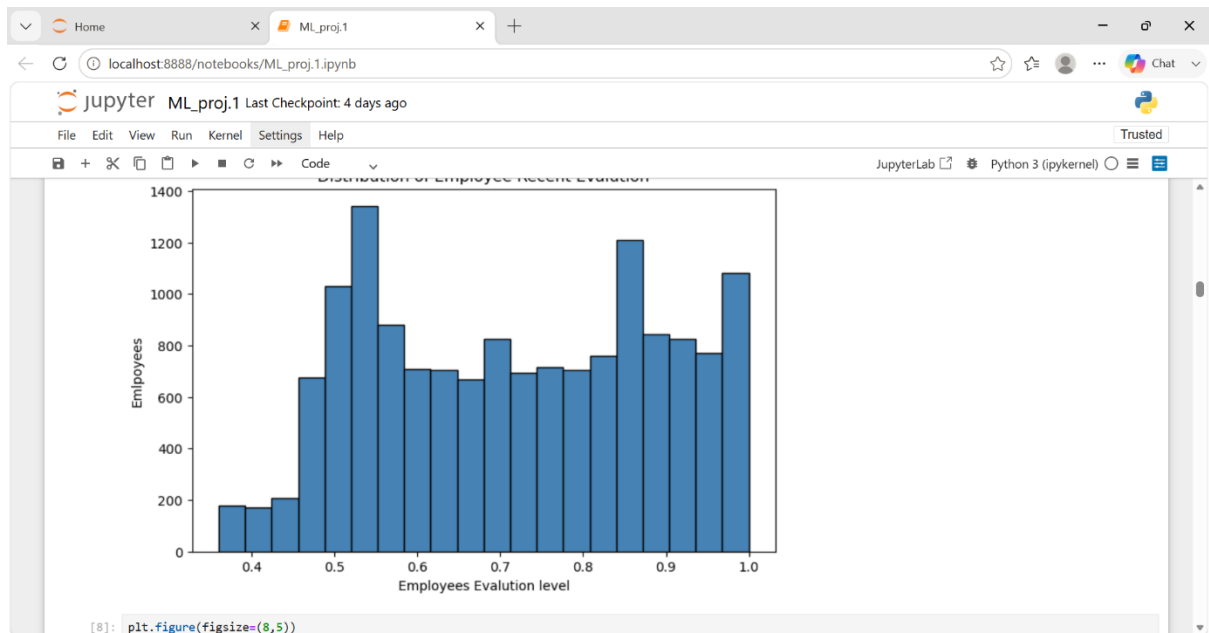
```
[15]: df.shape
```

```
[15]: (14999, 10)
```

```
[14]: df.describe()
```

```
[14]:
```

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years
count	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000
mean	0.612834	0.716102	3.803054	201.050337	3.498233	0.144610	0.238083	0.021268
std	0.248631	0.171169	1.232592	49.943099	1.460136	0.351719	0.425924	0.144281
min	0.090000	0.360000	2.000000	96.000000	2.000000	0.000000	0.000000	0.000000
25%	0.440000	0.560000	3.000000	156.000000	3.000000	0.000000	0.000000	0.000000
50%	0.640000	0.720000	4.000000	200.000000	3.000000	0.000000	0.000000	0.000000
75%	0.820000	0.870000	5.000000	245.000000	4.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	7.000000	310.000000	10.000000	1.000000	1.000000	1.000000



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JupyterLab Python 3 (ipykernel)

```
[22]: dummies=pd.get_dummies(category_col, drop_first=True).astype(int)
      dummies.head()
```

	sales_RandD	sales_accounting	sales_hr	sales_management	sales_marketing	sales_product_mng	sales_sales	sales_support	sales_technical	salary_low	salary_medium
0	0	0	0	0	0	0	1	0	0	1	0
1	0	0	0	0	0	0	1	0	0	0	1
2	0	0	0	0	0	0	1	0	0	0	1
3	0	0	0	0	0	0	1	0	0	1	0
4	0	0	0	0	0	0	1	0	0	1	0

```
[23]: numeric_data=pd.concat([numerical_col,dummies],axis=1)
      numeric_data.head()
```

	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_spend_company	Work_accident	left	promotion_last_5years	cluster	sales_RandD	sale
0	0.38	0.53	2	-0.882040	3	0	1	0	1	0	
1	0.80	0.86	5	1.220423	6	0	1	0	1	0	
2	0.11	0.88	7	1.420657	4	0	1	0	1	0	
3	0.72	0.87	5	0.439508	5	0	1	0	1	0	

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JupyterLab Python 3 (ipykernel)

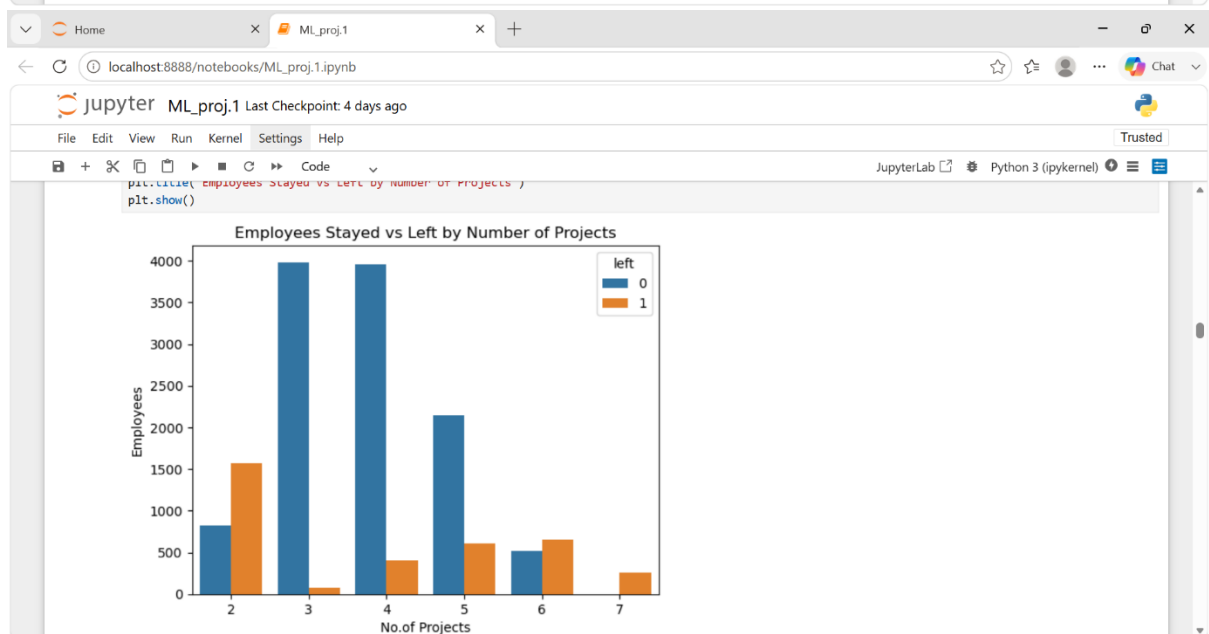
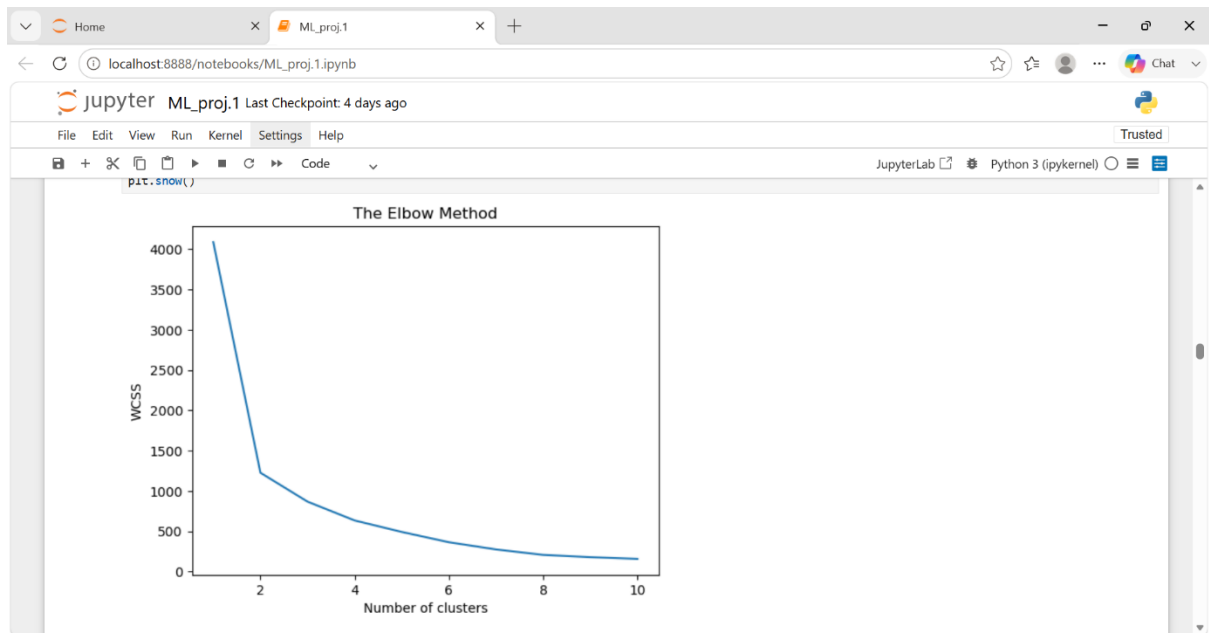
```
[33]: from sklearn.metrics import silhouette_score
      score=silhouette_score(x,y_kmeans)
      print(score)
      0.4339269204075329
```

```
[34]: df["cluster"] = y_kmeans
      df.groupby("cluster")[["satisfaction_level", "last_evaluation"]].mean()
```

	satisfaction_level	last_evaluation
cluster		
0	0.458195	0.681224
1	0.440098	0.718113
2	0.813334	0.739529

```
[35]: new_employee = [[0.75, 0.80, 0]]
      clus=model.predict(new_employee)
      print(clus)
      [2]
```

▼ Data Preprocessing and Class Imbalance Handling ¶



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JupyterLab Python 3 (ipykernel)

```
[9]: employees_left=pd.crosstab(df["number_project"],df["left"])
employees_left=pd.DataFrame(employees_left)
employees_left.columns=["Stayed(0)", "left(1)"]
employees_left
```

	Stayed(0)	left(1)
number_project		
2	821	1567
3	3983	72
4	3956	409
5	2149	612
6	519	655
7	0	256

```
[10]: sns.countplot(x="number_project", hue="left", data=df)
plt.xlabel("No. of Projects")
plt.ylabel("Employees")
plt.title("Employees Stayed vs Left by Number of Projects")
plt.show()
```

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JupyterLab Python 3 (ipykernel)

```
print(classification_report(y_test, log_reg_y_test_pred))
```

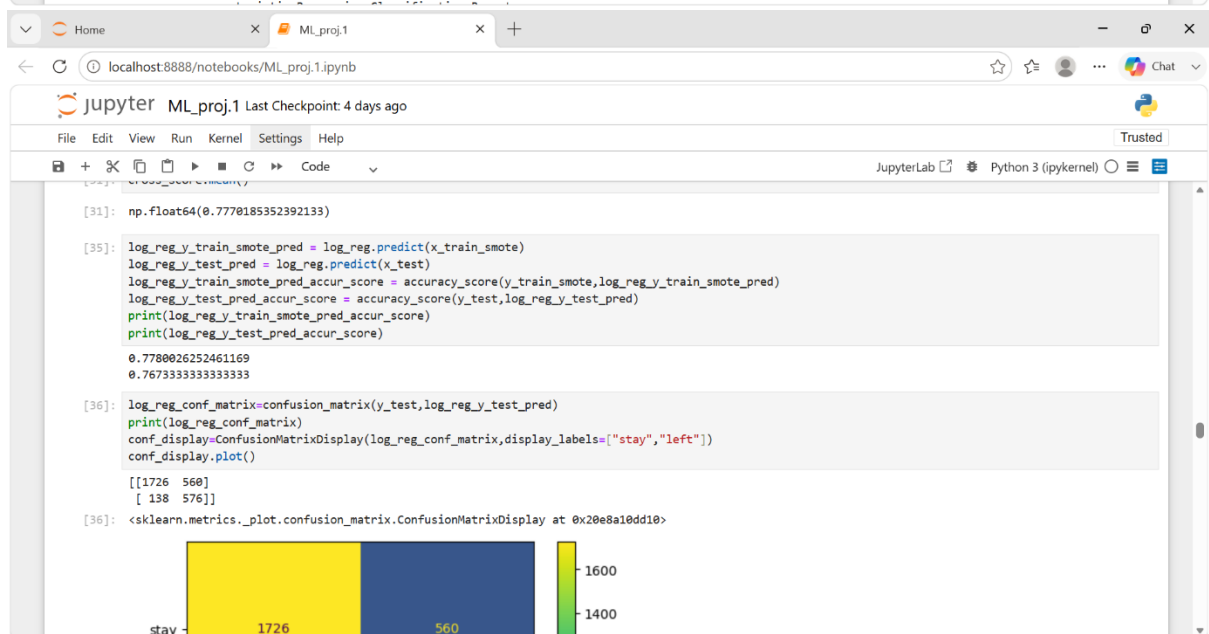
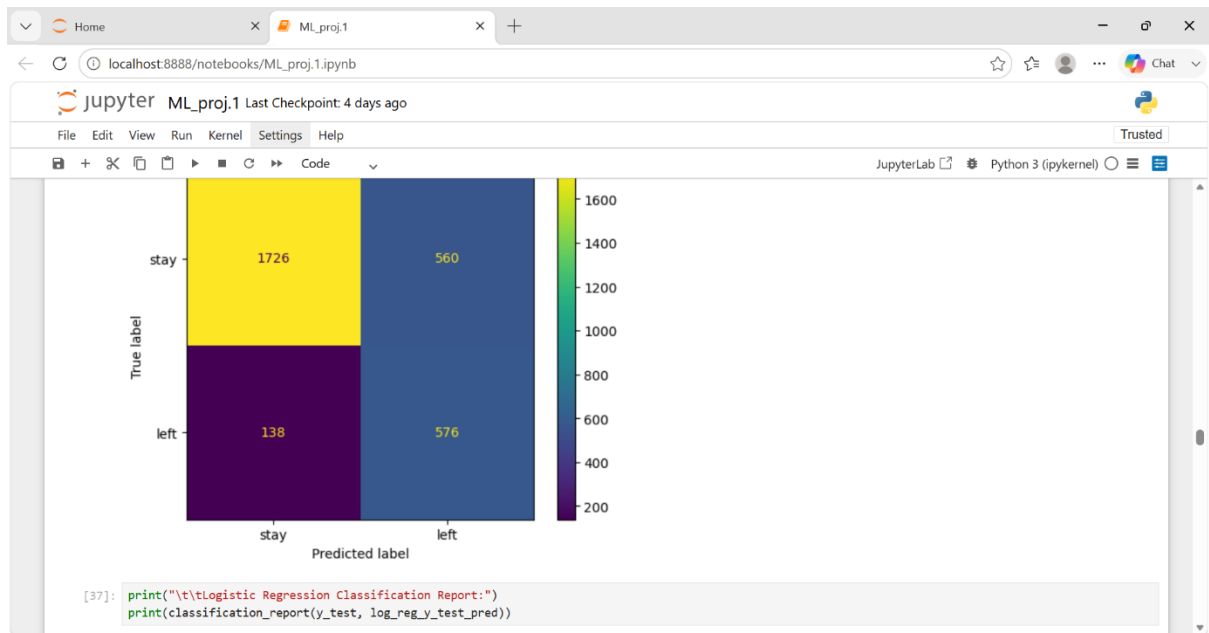
```
Logistic Regression Classification Report:
precision    recall  f1-score   support

      0       0.93      0.76      0.83      2286
      1       0.51      0.81      0.62       714

 accuracy          0.72          0.78          0.77      3000
 macro avg          0.72          0.78          0.73      3000
 weighted avg          0.83          0.77          0.78      3000
```

```
[38]: log_reg_prob=log_reg.predict_proba(x_test)[:,-1]
log_reg_df=pd.DataFrame({"actual_values":y_test,
                        "pred_values":log_reg_y_test_pred,
                        "probability":log_reg_prob
                        })
log_reg_df.head()
```

	actual_values	pred_values	probability
10627	0	0	0.078950
2703	0	1	0.826821
6059	0	0	0.491286
3258	0	0	0.180691



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JupyterLab Python 3 (ipykernel)

```
[27]: y_train.value_counts()

[27]: left
      0    9142
      1    2857
      Name: count, dtype: int64

[28]: y_train_smote.value_counts()

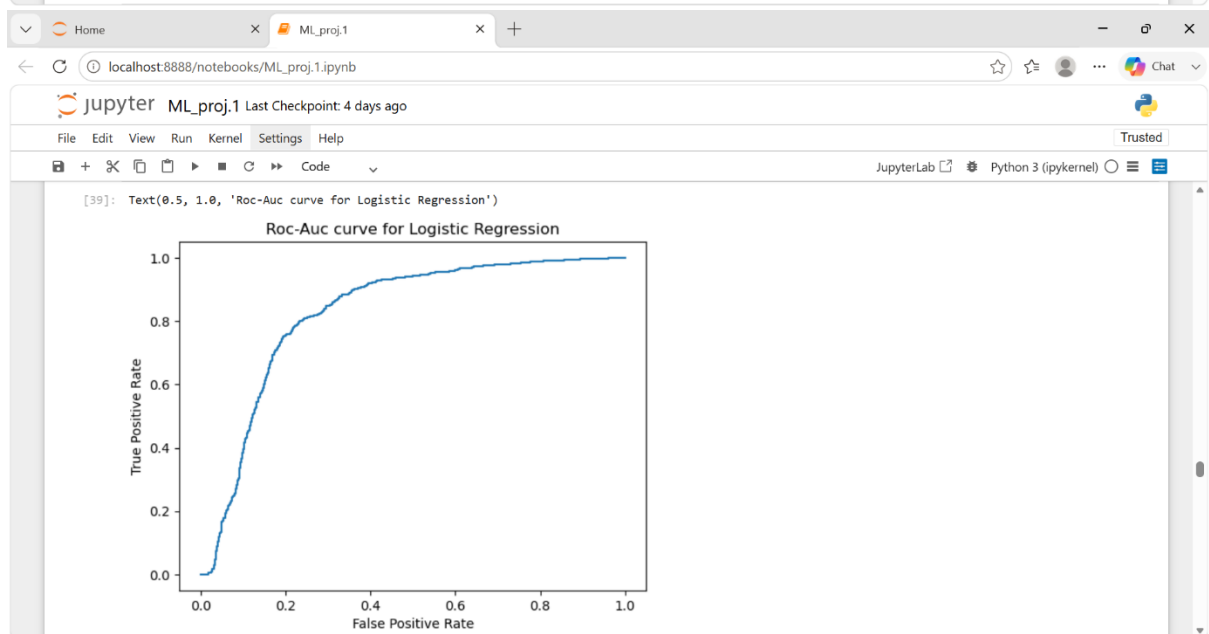
[28]: left
      0    9142
      1    9142
      Name: count, dtype: int64

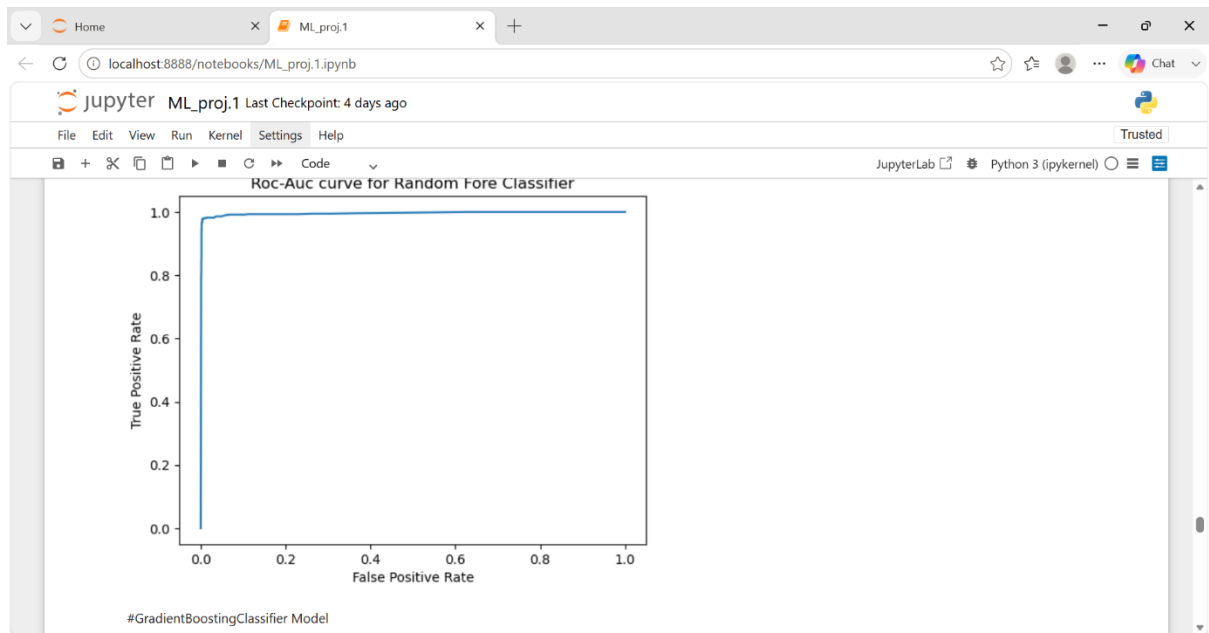
#LogisticRegression Model

[29]: from sklearn.linear_model import LogisticRegression
      from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, ConfusionMatrixDisplay
      log_reg=LogisticRegression()
      log_reg.fit(x_train_smote,y_train_smote)

[29]: LogisticRegression

[30]: cross_score=cross_val_score(log_reg,x_train_smote,y_train_smote,cv=5,scoring="accuracy")
```





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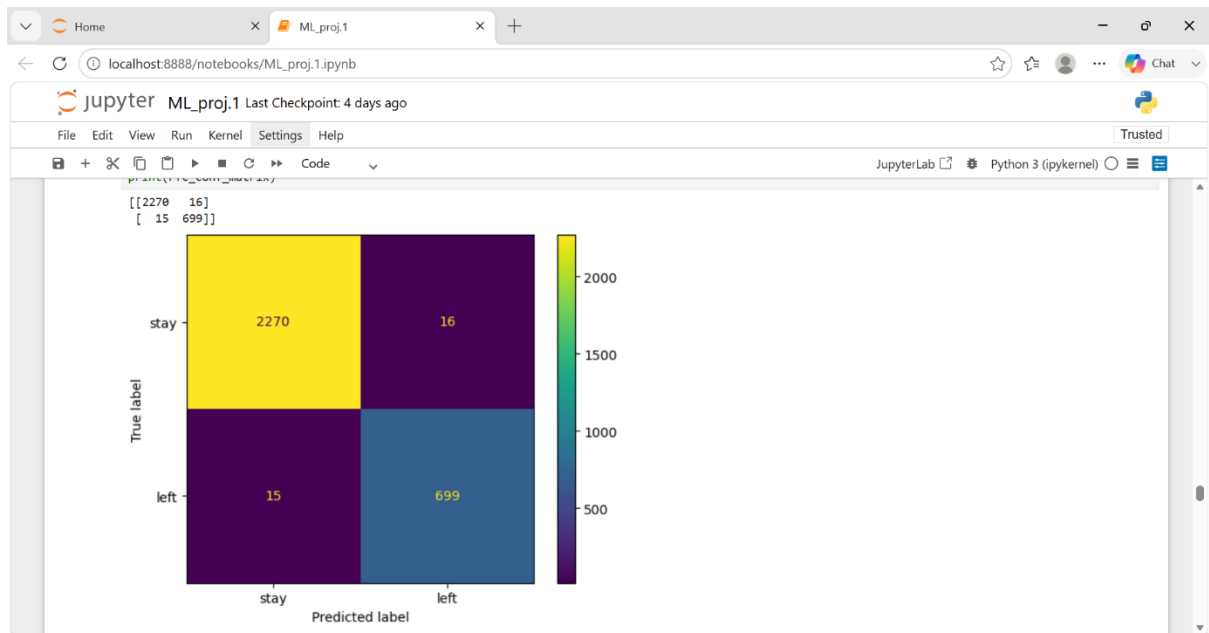
JupyterLab Python 3 (ipykernel)

```
[48]: rfc_cls_report=classification_report(y_test, rfc_y_test_pred)
print(rfc_cls_report)
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	2286
1	0.98	0.98	0.98	714
accuracy			0.99	3000
macro avg	0.99	0.99	0.99	3000
weighted avg	0.99	0.99	0.99	3000

```
[49]: rfc_prob=RFC.predict_proba(x_test)[:,:1]
rfc_prob = RFC.predict_proba(x_test)[:,:1]
rfc_df=pd.DataFrame({"actual_values":y_test,
                    "pred_values":rfc_y_test_pred,
                    "probability":rfc_prob
                    })
rfc_df.head()
```

	actual_values	pred_values	probability
10627	0	0	0.00
2703	0	1	0.64
6059	0	0	0.06



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JupyterLab Python 3 (ipykernel)

```
[43]: from sklearn.ensemble import RandomForestClassifier
      RFC_model = RandomForestClassifier(n_estimators=100)
      CV_score = cross_val_score(RFC_model, X_train_smote, y_train_smote, cv=5, scoring="accuracy")

[44]: CV_score.mean()

[44]: np.float64(0.9839750839043134)

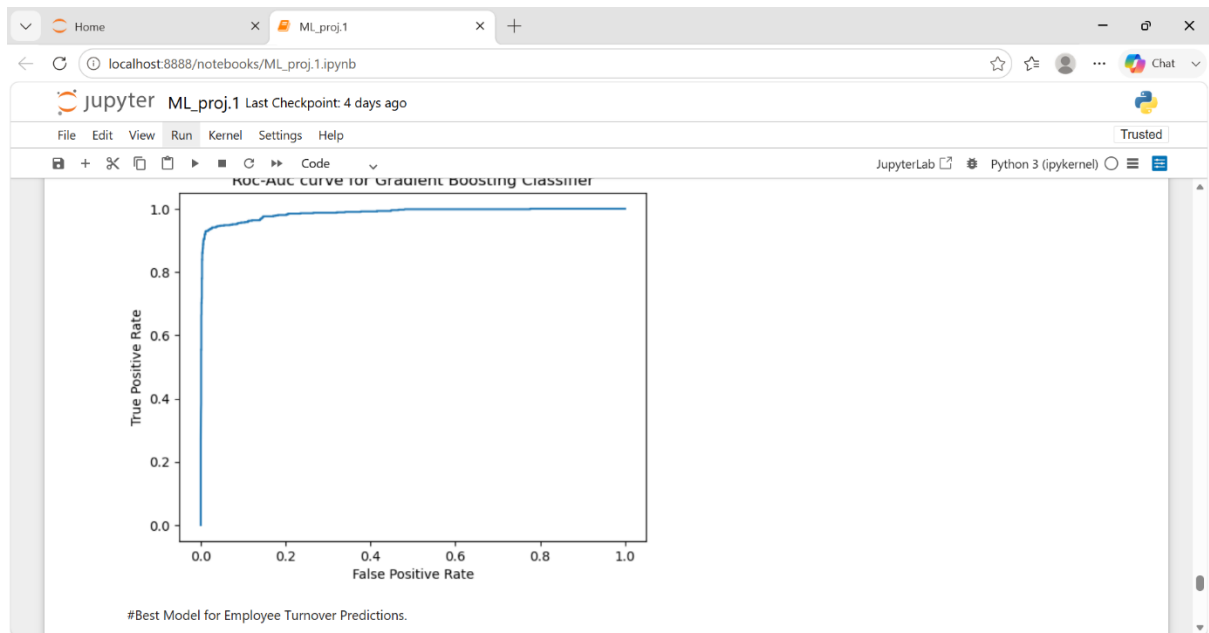
[45]: RFC_model.fit(X_train_smote, y_train_smote)

[45]: RandomForestClassifier
      RandomForestClassifier()

[46]: rfc_y_train_smote_pred = RFC_model.predict(X_train_smote)
      rfc_y_test_pred = RFC_model.predict(X_test)
      rfc_y_train_smote_acc_score = accuracy_score(y_train_smote, rfc_y_train_smote_pred)
      rfc_y_test_acc_score = accuracy_score(y_test, rfc_y_test_pred)
      print(rfc_y_train_smote_acc_score)
      print(rfc_y_test_acc_score)

1.0
0.9896666666666667

[47]: rfc_conf_matrix = confusion_matrix(y_test, rfc_y_test_pred)
      conf_display = ConfusionMatrixDisplay(rfc_conf_matrix, display_labels=["stay", "left"])
      conf_display.plot()
```



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JupyterLab Python 3 (ipykernel)

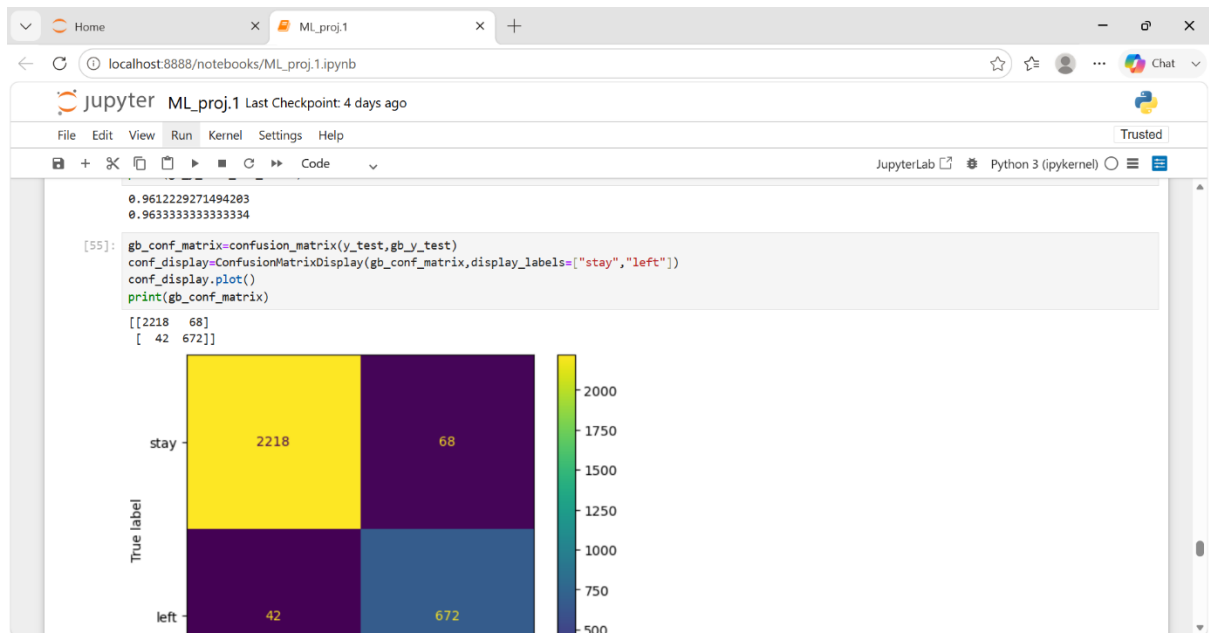
```
[56]: GB_cls_report=classification_report(y_test, gb_y_test)
print(GB_cls_report)
```

	precision	recall	f1-score	support
0	0.98	0.97	0.98	2286
1	0.91	0.94	0.92	714
accuracy			0.96	3000
macro avg	0.94	0.96	0.95	3000
weighted avg	0.96	0.96	0.96	3000

```
[57]: gb_prob = GB_model.predict_proba(x_test)[:,:1]
gb_df=pd.DataFrame({"actual_values":y_test,
                    "pred_values":gb_y_test,
                    "probability":gb_prob
                    })
gb_df.head()
```

```
[57]:
```

	actual_values	pred_values	probability
10627	0	0	0.006421
2703	0	1	0.935316
6059	0	0	0.091130
3258	0	0	0.012963



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JupyterLab Python 3 (ipykernel)

GradientBoostingClassifier Model

```
[51]: from sklearn.ensemble import GradientBoostingClassifier
      GB_model=GradientBoostingClassifier(n_estimators=100, random_state=7)
      GB_CV_score=cross_val_score(GB_model,x_train_smote,y_train_smote,cv=5,scoring="accuracy")
      GB_CV_score

[51]: array([0.96035001, 0.95652174, 0.9630845 , 0.95843588, 0.95650985])

[52]: GB_CV_score.mean()

[52]: np.float64(0.9589803943039008)

[53]: GB_model.fit(x_train_smote,y_train_smote)

[53]: GradientBoostingClassifier
      GradientBoostingClassifier(random_state=7)

[54]: gb_y_train_smote_pred=GB_model.predict(x_train_smote)
      gb_y_test=GB_model.predict(x_test)
      gb_y_train_acc_score=accuracy_score(y_train_smote,gb_y_train_smote_pred)
      gb_y_test_acc_score=accuracy_score(y_test, gb_y_test)
      print(gb_y_train_acc_score)
      print(gb_y_test_acc_score)
```

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JupyterLab Python 3 (ipykernel)

```
results["Risk_Zone"] = results[
    "Turnover_Probability"
].apply(risk_zone)

results.head()
```

[68]:

	Employee_ID	Turnover_Probability	Risk_Zone
0	10627	0.00	Safe Zone (Green)
1	2703	0.64	Medium-Risk Zone (Orange)
2	6059	0.06	Safe Zone (Green)
3	3258	0.01	Safe Zone (Green)
4	4565	0.00	Safe Zone (Green)

[]:

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JupyterLab Python 3 (ipykernel)

Best Model for Employee Turnover Predictions.

```
[59]: results = pd.DataFrame({
    "Employee_ID": x_test.index,
    "Turnover_Probability": rfc_prob
})

results.head()
```

[59]:

	Employee_ID	Turnover_Probability
0	10627	0.00
1	2703	0.64
2	6059	0.06
3	3258	0.01
4	4565	0.00

```
[60]: def risk_zone(prob):
    if prob < 0.20:
        return "Safe Zone (Green)"
```